

Lecture 3 Notes: Plato's Rationalism, and Aristotle

Philosophy of Mathematics

Central Theme

Plato and Aristotle offer two rival ways to explain the exactness and objectivity of mathematics.

Plato separates mathematics from the physical world.
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Aristotle embeds mathematics in the physical world.

The guiding contrast:

Plato	Aristotle
explains exactness	explains applicability
makes application mysterious	makes exactness mysterious

1. Greek Mathematics: Exactness and Proof

Greek mathematics introduced a distinctive demand for exactitude and rigorous proof.

The story of doubling the altar illustrates this demand. A practical problem becomes a mathematical problem: not approximately double the altar, but exactly double it.

The resulting problem is the problem of doubling the cube:

Given a line segment s , construct x such that $x^3 = 2s^3$.

Similar ancient problems include:

trisecting an angle

squaring the circle

In each case, approximation was not enough. The Greek mathematicians wanted exact construction and proof.

Teaching Point

Greek mathematics is philosophically important because it treats exact proof, not practical approximation, as the standard of mathematical success.

Whiteboard

Practical problem $\longrightarrow$ exact construction $\longrightarrow$ proof

$2.0004 \neq 2$ for the mathematician

2. Plato's World of Being

Plato begins from a gap between imperfect physical things and perfect intelligible ideals.

We encounter beautiful things, just actions, and circular objects. But we do not encounter Beauty itself, Justice itself, or the perfect Circle in the physical world.

Plato explains this by distinguishing two realms:

World of Becoming	World of Being
physical	intelligible
changing	eternal
imperfect	perfect
known by senses	known by thought

The Forms belong to the world of Being. Physical objects resemble, participate in, or approximate the Forms.

Teaching Point

Plato's metaphysics explains why physical things are imperfect: they are not the standards themselves, but only imperfect participants in those standards.

Whiteboard

beautiful things $\neq$ Beauty itself

just actions $\neq$ Justice itself

drawn circles $\neq$ Circle itself

3. Plato's Epistemology: Thought over Sense

For Plato, the senses give us access to the world of Becoming. Reason gives us access to the world of Being.

This is why mathematics is central. Mathematics trains the soul to turn away from the changing world of sense and toward eternal, intelligible reality.

The *Meno* provides an important example. Socrates guides a slave to a geometrical truth without directly teaching it to him. Plato uses this as evidence for recollection: what we call learning is really remembering.

Even if we do not accept the doctrine of recollection, the main point remains: mathematical knowledge is not ordinary sensory knowledge.

Teaching Point

For Plato, mathematics is a bridge from sensory experience to rational insight.

Whiteboard

sense perception $\longrightarrow$ Becoming

reason $\longrightarrow$ Being

mathematics = training for ascent

4. Geometry as Plato's Model

Geometry gives Plato a clear example of the gap between the physical and the intelligible.

A theorem of geometry concerns perfect points, lines, and circles. But no physical drawing contains a perfect point, a breadthless line, or a perfect circle.

Consider the theorem:

A tangent to a circle intersects the circle at exactly one point.

In a physical drawing, the line and the circle overlap in a small region. That does not disconfirm the theorem. Instead, for Plato, it shows that the theorem is not about the drawing.

Teaching Point

A diagram is not the object of geometry. It is an image or aid that helps the mind grasp the ideal geometrical object.

Whiteboard

drawn circle $\neq$ Circle itself

drawn tangent $\neq$ ideal tangent

diagram = image, not object

5. Plato's Divided Line

In the *Republic*, Plato uses the divided line to represent levels of reality and cognition.

The Good
Forms
Mathematical objects
Physical objects
Reflections/images

Mathematical objects are above physical objects but below the Forms. They belong to the world of Being, but mathematicians still rely on hypotheses and diagrams.

Teaching Point

Mathematics is elevated above ordinary sensory knowledge, but it is not yet the highest form of knowledge. It prepares the soul for dialectic and the grasp of the Forms.

Whiteboard

GOOD
FORMS
MATHEMATICAL OBJECTS
PHYSICAL OBJECTS
IMAGES

mathematics = middle step upward

6. Plato's Criticism of Mathematical Language

Plato criticizes ordinary geometrical language because it sounds dynamic and constructive.

Geometers say:

draw a line construct a triangle move a figure

But if the true objects of geometry are eternal and unchanging, then they are not literally drawn, constructed, or moved.

Teaching Point

Plato is an example of philosophy-first. His metaphysics of mathematical objects leads him to criticize ordinary mathematical language.

Whiteboard

Mathematical objects are eternal and unchanging.

∴ “draw a line” is philosophically misleading.

7. Plato on Arithmetic and Pure Units

Plato applies the same contrast to arithmetic.

Ordinary arithmetic counts physical things:

two cows two armies two shoes

But these units are unequal. Two shoes differ in shape. Two cows differ in size. Two armies differ enormously.

Philosophical arithmetic concerns pure units. Each unit is precisely equal to every other unit.

Ordinary arithmetic	Philosophical arithmetic
physical units	pure units
unequal	equal
practical counting	knowledge of number itself

Teaching Point

For Plato, arithmetic, like geometry, is strictly about intelligible objects rather than physical ones.

Whiteboard

ordinary two = two unequal physical things

philosophical two = two pure equal units

8. Mathematics on Plato

The chapter then reverses direction. It asks not only what Plato thought about mathematics, but how mathematics shaped Plato's philosophy.

Socrates and Plato differ sharply.

Socrates	Plato
public questioning	elite training
ethics and politics	mathematics and metaphysics
fallible dialectic	rigorous demonstration
philosophy for everyone	philosophy for Guardians

Socrates uses questioning to expose confusion. Plato, influenced by mathematics, wants certainty, proof, and disciplined ascent toward first principles.

Teaching Point

Greek mathematics helped transform philosophy into rationalism: the search for certain knowledge through reason.

Whiteboard

Socratic questioning → fallible testing

Mathematical proof → certainty

Plato: philosophy should imitate mathematics.

9. Mathematics and the Guardians

In the *Republic*, the future Guardians study mathematics for ten years, from roughly age 20 to age 30.

This training is not mainly practical. It is spiritual and intellectual. Mathematics turns the soul away from the world of change and toward reality.

After mathematics, the Guardians study dialectic. Dialectic aims at unhypothetical first principles and ultimately at the Form of the Good.

Teaching Point

For Plato, mathematics is not the final goal of philosophy. It is preparation for dialectic.

Whiteboard

Mathematics $\longrightarrow$ Dialectic $\longrightarrow$ The Good

hypotheses $\longrightarrow$ first principles

10. Aristotle's Rejection of Separate Forms

Aristotle rejects Plato's separate world of Forms.

For Plato, Forms exist apart from physical things. For Aristotle, Forms exist in things.

Plato	Aristotle
Forms separate from things	Forms in things
Being apart from Becoming	form embedded in nature
mathematical objects separate	mathematics studies aspects of things

Beauty is not a separate entity apart from beautiful things. Beauty is what beautiful things have in common.

Likewise, mathematical objects do not need to belong to a separate realm.

Teaching Point

Aristotle brings Forms down into the world. This changes the philosophy of mathematics: mathematics must now be understood in relation to perceptible objects.

Whiteboard

Plato: Form apart from things

Aristotle: form in things

11. Aristotle on Mathematical Abstraction

Aristotle says that the mathematician studies physical objects, but not as physical.

A geometer may consider a brass sphere, but not as brass. She considers it as spherical.

brass sphere $\longrightarrow$ ignore brass $\longrightarrow$ sphere

An arithmetician may consider a flock of sheep, but not as sheep. She considers them as units.

five sheep $\longrightarrow$ ignore sheepness $\longrightarrow$ 5

The separation is not metaphysical. It is separation in thought.

Teaching Point

For Aristotle, mathematical abstraction does not require a separate mathematical realm. The mathematician focuses on certain features of physical things and ignores others.

Whiteboard

Mathematics studies physical things

but not qua physical.

separation in thought $\neq$ separation in reality

12. Two Interpretations of Aristotle

The chapter discusses two interpretations of Aristotle's account.

Abstractionist interpretation	Fictionalist interpretation
mathematical objects are abstracted	mathematical objects are useful fictions
objects exist as forms of things	strictly, no separate objects
realism in ontology	anti-realism in ontology
realism in truth-value	realism in truth-value

On the abstractionist view, mathematical objects exist as abstractions from physical objects.

On the fictionalist view, we do not literally create or discover separate mathematical objects. We pretend there are such objects because doing so is useful and harmless.

Teaching Point

Both interpretations try to preserve the objectivity of mathematics without Plato's separate world of Being.

Whiteboard

Aristotle: objective mathematics without separate Forms?

Abstraction | Useful fiction

13. Frege's Criticism of Abstraction

The chapter uses Frege to raise a problem for abstraction.

Suppose we try to obtain the number of a group of objects by abstracting away their differences. If abstraction removes all differences, then we cannot count them as many.

If the objects become completely identical, then we have only one object.

No difference $\Rightarrow$ no counting beyond one

But if the differences remain, then abstraction has not produced pure identical units.

Teaching Point

Abstraction is difficult to explain. If we abstract too much, we lose multiplicity. If we abstract too little, we do not get pure mathematical objects.

Whiteboard

Too much abstraction $\Rightarrow$ one indistinguishable thing

Too little abstraction $\Rightarrow$ ordinary physical differences remain

14. Aristotle's Advantage: Applicability

Aristotle has an advantage over Plato: he explains why mathematics applies to the physical world.

If mathematics studies aspects of physical things, then it is no mystery that mathematics applies to physical things.

Mathematics studies physical things under abstraction.

$\therefore$ Mathematics applies to physical things.

Unlike Plato, Aristotle does not need to explain how a separate mathematical realm is connected to the physical realm.

Teaching Point

Aristotle makes applicability straightforward by refusing to separate mathematics from the physical world.

Whiteboard

Aristotle's advantage:

No gap between mathematics and nature.

Application is built in.

15. Aristotle's Problem: Exactness

Aristotle's problem is the reverse of Plato's.

Physical objects are imperfect. A brass sphere is not perfectly spherical. A drawn line has thickness. A physical tangent overlaps a physical circle over a region.

So if mathematics is about physical things, or abstractions from physical things, how can mathematics be exact?

imperfect physical object $\longrightarrow?$ perfect mathematics

This is the central difficulty for Aristotle.

Teaching Point

Plato explains exactness but struggles with application. Aristotle explains application but struggles with exactness.

Whiteboard

Plato	Aristotle
exactness explained	application explained
application mysterious	exactness mysterious

16. Limits of Aristotle's Approach

Aristotle's approach works best for mathematics close to ordinary physical experience: geometry, counting, ratios, and perhaps parts of analysis.

But modern mathematics extends far beyond direct abstraction from physical objects.

Examples:

complex analysis

functional analysis

point-set topology

axiomatic set theory

A modern Aristotelian must explain how these areas relate to the physical world, if they do at all.

Teaching Point

Aristotle anticipates empiricist approaches to mathematics, but his view faces pressure from highly abstract modern mathematics.

Whiteboard

Counting sheep and brass spheres

$\neq$

set theory, topology, functional analysis

Conclusion: Plato and Aristotle as Opposing Models

The chapter presents Plato and Aristotle as the first major contrast in philosophy of mathematics.

Plato	Aristotle
mathematics belongs to Being	mathematics abstracts from Becoming
objects are separate and ideal	forms are in things
knowledge by reason/recollection	knowledge through abstraction
exactness is secure	applicability is secure
application is difficult	exactness is difficult

Final Framing

To teach this chapter well, present it as a foundational opposition.

Plato and Aristotle are not merely historical figures. They introduce two permanent tendencies in the philosophy of mathematics.

Platonist tendency: protect exactness by separating mathematics from nature.
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Aristotelian tendency: protect applicability by embedding mathematics in nature.
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Later philosophies of mathematics repeatedly return to this tension.

Discussion Questions

1. Why did Greek mathematicians care so much about exactness rather than approximation?
2. Does the tangent theorem concern the drawn circle or the ideal circle?
3. What role do diagrams play in geometrical proof?
4. Is Plato right that mathematical knowledge is independent of the senses?
5. Why does Plato think mathematics prepares the soul for philosophy?
6. Is philosophical arithmetic different from ordinary counting?
7. Can Aristotle explain exact mathematical truth without a separate mathematical realm?
8. Is abstraction a plausible source of mathematical objects?
9. Does Aristotle explain the applicability of mathematics better than Plato?
10. Which is more important for a philosophy of mathematics: exactness or applicability?

Key Terms

World of Becoming Plato's name for the changing, physical, sensory world.

World of Being Plato's name for the eternal, intelligible realm of Forms and mathematical objects.

Form A perfect, intelligible standard such as Beauty itself, Justice itself, or Circle itself.

Rationalism The view that reason is a fundamental source of knowledge.

Recollection Plato's doctrine that learning is remembering what the soul already knew.

Divided line Plato's image in the *Republic* representing levels of reality and cognition.

Mathematical object An object studied by mathematics, such as a number, point, line, circle, or sphere.

Pure unit A unit in philosophical arithmetic, precisely equal to every other unit.

Dialectic For Plato, the highest philosophical method, aimed at first principles and the Forms.

Abstraction The process of focusing on some features of things while ignoring others.

Qua Latin for "as"; to study something qua mathematical is to study it under its mathematical aspect.

Fictionalism The view that mathematical objects are useful fictions rather than literally existing objects.

Applicability problem The problem of explaining why mathematics applies to the physical world.